Logic Puzzles!

Based on the stories by Raymond Smullyan

Kitten or Tiger

There are two doors with signs. Choose wisely.

Trial 1

One sign is true, but the other one is false.

In this room there is a kitten, and in the other room there is a tiger

I

In one of these rooms there is a kitten, and in one of these rooms there is a tiger

II

SOLUTION

EXPLANATION

In this room there is a kitten, and in the other room there is a tiger

Suppose TRUE.
Then I = kitten and II = tiger

Suppose FALSE.
Then I = tiger or II = kitten

I



In one of these rooms there is a kitten, and in one of these rooms there is a tiger

Suppose FALSE.
Then I = II

Suppose TRUE.
Then I = tiger and II = kitten

II

TRUE

In this room there is a kitten, and in the other room there is a tiger

I	II
Tiger	Tiger
Tiger	Kitten
Kitten	Kitten
Kitten	Tiger

I

SOLUTION

EXPLANATION

FALSE

In one of these rooms there is a kitten, and in one of these rooms there is a tiger

I	II
Tiger	Tiger
Tiger	Kitten
Kitten	Kitten
Kitten	Tiger

II

FALSE

In this room there is a kitten, and in the other room there is a tiger

I	II
Tiger	Tiger
Tiger	Kitten
Kitten	Kitten
Kitten	Tiger

I

SOLUTION EXPLANATION

TRUE

In one of these rooms there is a kitten, and in one of these rooms there is a tiger

I	II
Tiger	Tiger
Tiger	Kitten
Kitten	Kitten
Kitten	Tiger

II

Trial 2

They are either both true or both false.

At least one of these rooms contains a kitten

I

A tiger is in the other room

II

Trial 3

They are either both true or both false.

Either a tiger is in this room or a kitten is in the other room

I

A kitten is in the other room

II

Trial 4

If a kitten is in room I, then the sign is true but if a tiger is in it, the sign is false.

For room II, the situation is the opposite.

Both rooms contain kittens

I

Both rooms contain kittens

II

Trial 5

Same rules as Trial 4

At least one room contains a kitten

I

The other room contains a kitten

II

Trial 6

Same rules as Trial 5

It makes no difference which room you pick

I

There is a kitten in the other room

II

Trial 7

Same rules as Trial 6

It does make a difference which room you pick

I

You are better off choosing the other room

II

Trial 8

Same rules as Trial 7 but signs are not labeled

This room contains a tiger

Both rooms contain tigers